Kinvecta

A KINVECTA SERIES

How Large Language Models Work

From math to transformers

EPISODE 02 · VECTORS



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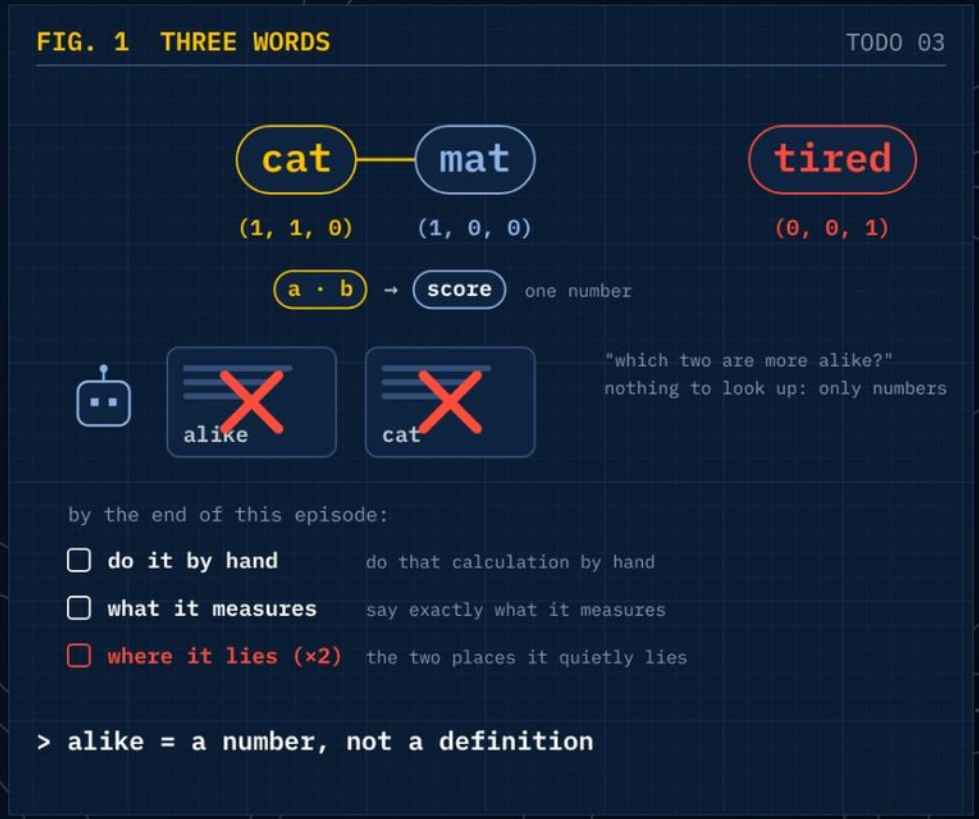
HOW LARGE LANGUAGE MODELS WORK · EPISODE 2

Cat. Mat. Tired. Which two are alike?

A model has no definition of “cat.” It has numbers, and one calculation.

13-minute lesson

Companion notebook



MARK

Numbers as geometry

Four parts, one calculation, and a notebook with every number.

01 What a vector is: a list, a point, an arrow

02 The dot product and the norm

03 Cosine similarity: the score for “alike”

04 Where it breaks: an analogy and a myth



Companion notebook: “Episode 02 - Vectors” (Colab), linked in the description. Sources on every slide.

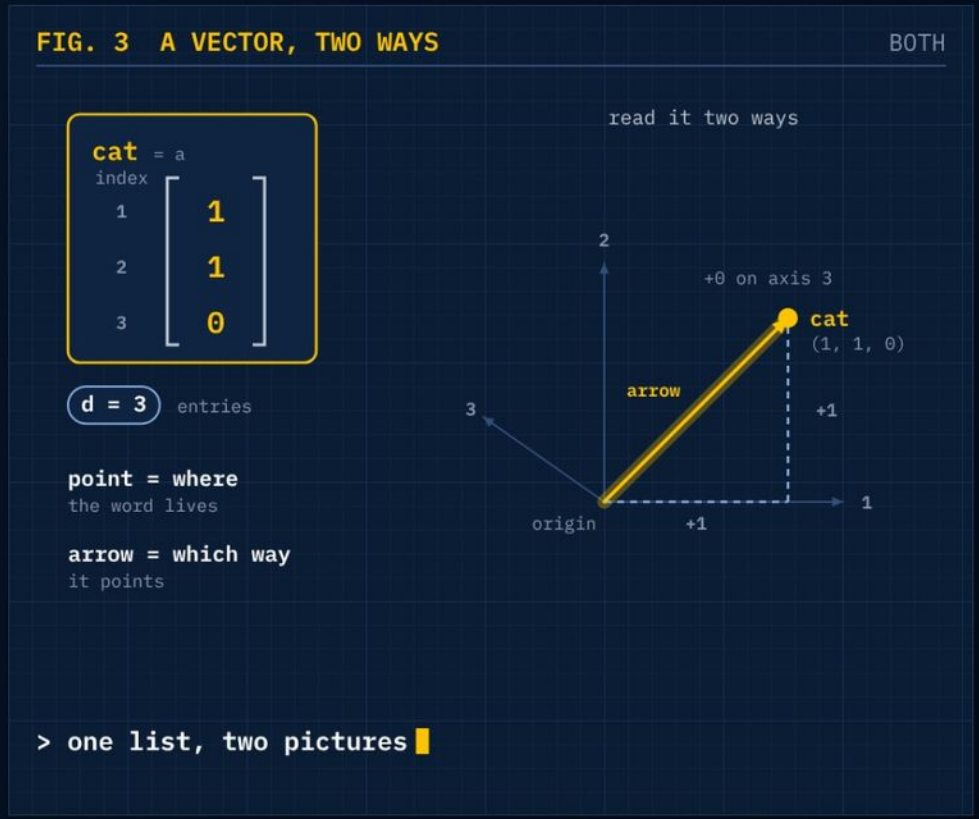
A vector is an ordered list

Read the same numbers two ways.

01 Dimension d: how many numbers

02 As a point: where the word lives

03 As an arrow: which way it points



Toy vectors chosen by hand for this episode (axes named by Mark); illustrative, not learned values.

PART 1 · THREE WORDS

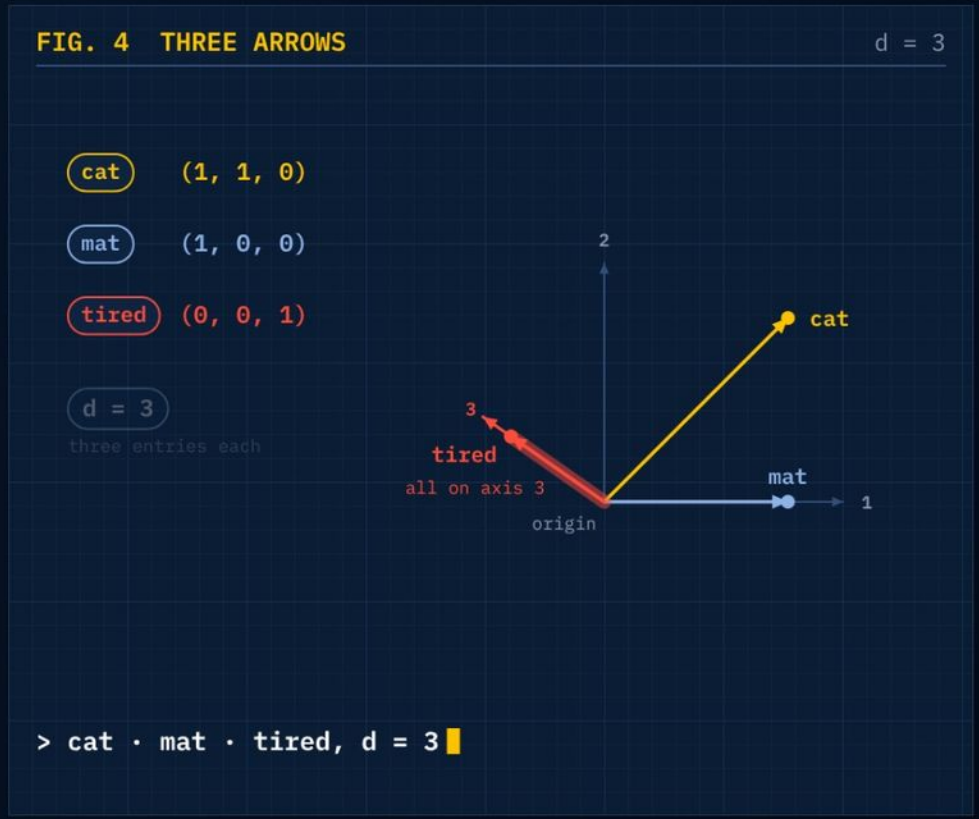
Three words, three arrows

`cat = (1, 1, 0) · mat = (1, 0, 0) · tired = (0, 0, 1)`

01 mat: one step along axis 1

02 tired: all of it on axis 3

03 d = 3 for every word



Toy vectors chosen by hand for this episode (axes named by Mark); illustrative, not learned values.

PART 1 · ONE HONEST WARNING

Real vectors are long

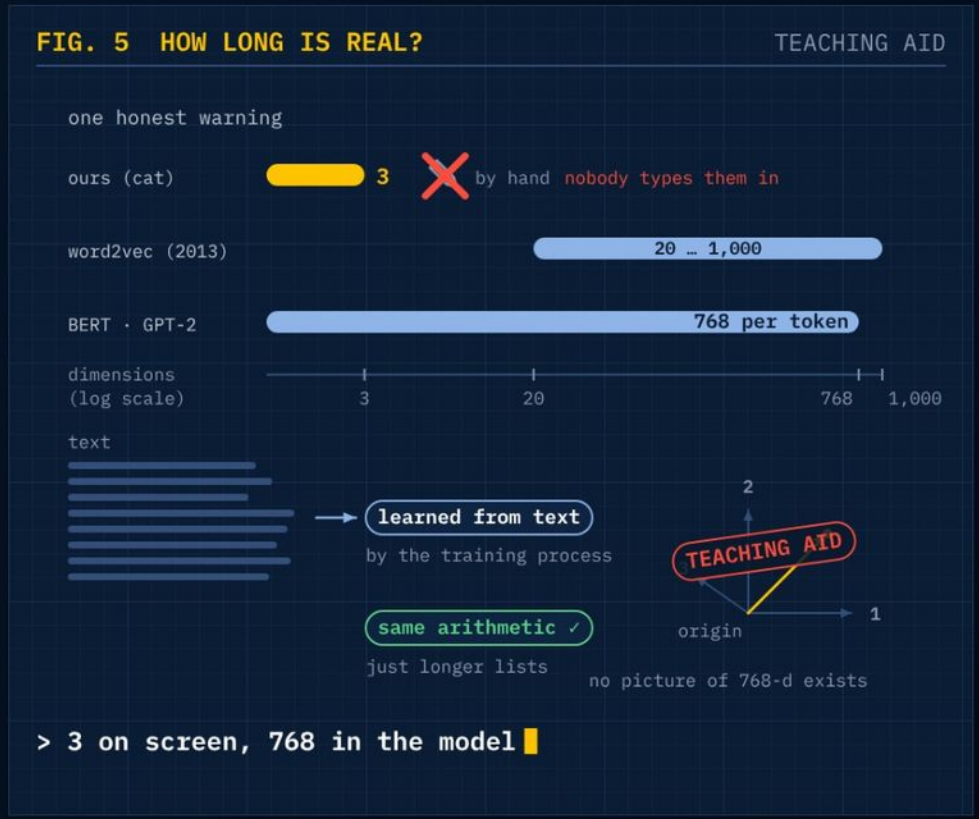
Three numbers fit on a screen. Models use hundreds.

01 word2vec tested 20 to 1,000 dimensions

02 BERT and GPT-2: 768 numbers per token

03 Learned from text, never typed in

04 Every picture today is a teaching aid



Mikolov, Chen, Corrado & Dean, “Efficient Estimation of Word Representations in Vector Space,” arXiv:1301.3781 (2013). Timkey & van Schijndel, “All Bark and No Bite: Rogue Dimensions in Transformer Language Models,” arXiv:2109.04404 (2021).

PART 2 · THE DOT PRODUCT

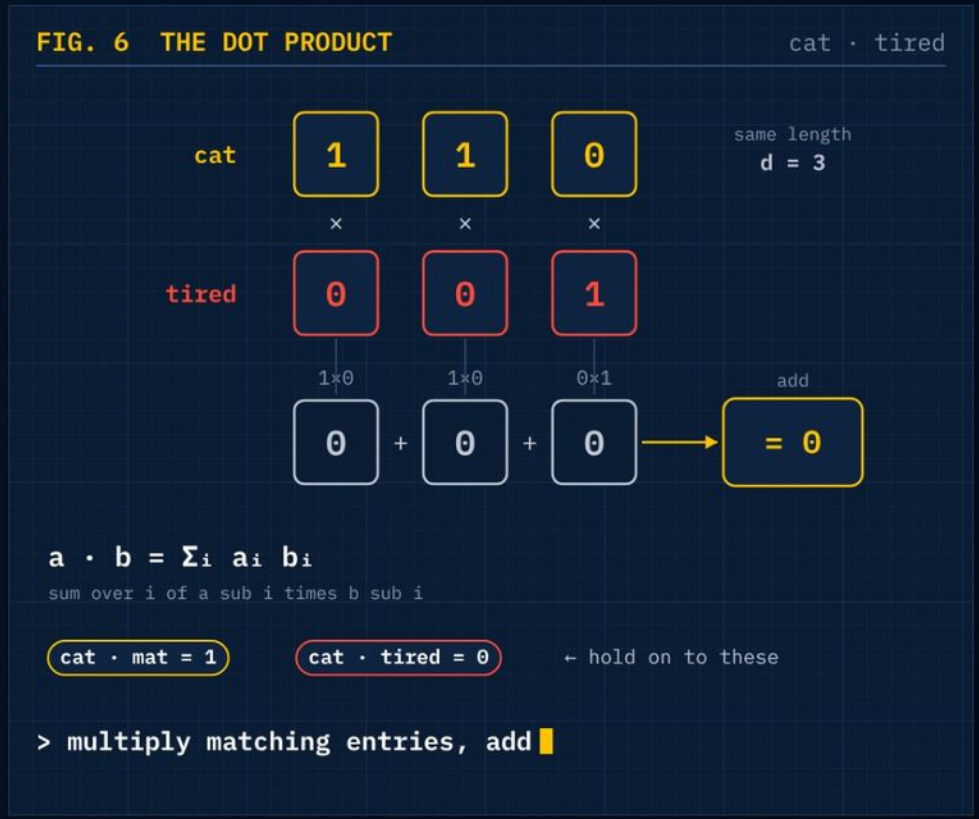
Multiply matching entries, then add

$$a \cdot b = \Sigma_i \, a_i \, b_i$$

01 cat · mat = 1·1 + 1·0 + 0·0 = 1

02 cat · tired = 1·0 + 1·0 + 0·1 = 0

03 Two numbers to hold on to



Wyss-Gallifent, Math 241 lecture notes §11.3 (Dot Product), University of Maryland. Toy vectors chosen by hand for this episode (axes named by Mark); illustrative, not learned values.

PART 2 · WHAT IT MEASURES

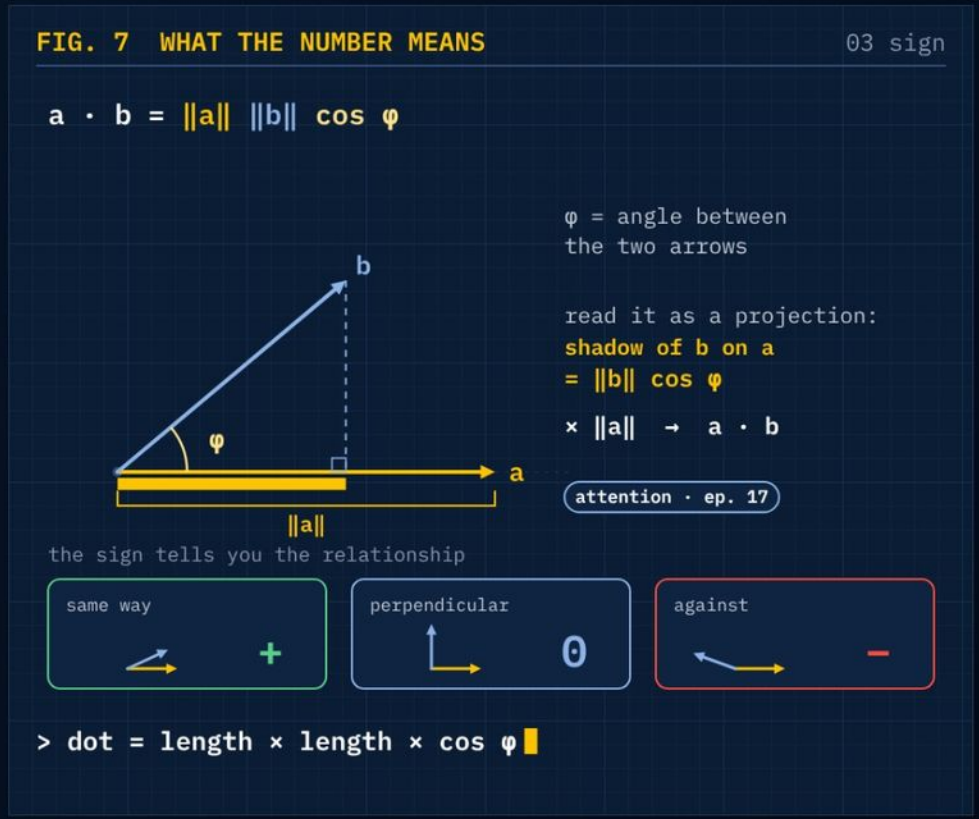
Length × length × cos φ

a · b = ‖a‖ ‖b‖ cos φ. The same number, read as geometry.

01 Projection: how far one arrow reaches along the other

02 Same way: positive · perpendicular: 0 · against: negative

03 Episode 17: attention is this one sentence



3Blue1Brown, “Dot products and duality,” Essence of linear algebra (2016). Wyss-Gallifent, Math 241 lecture notes §11.3 (Dot Product), University of Maryland. Alignment reading: IBM, “What is a transformer model?” (ibm.com/think).

PART 2 · A PICTURE TO KEEP

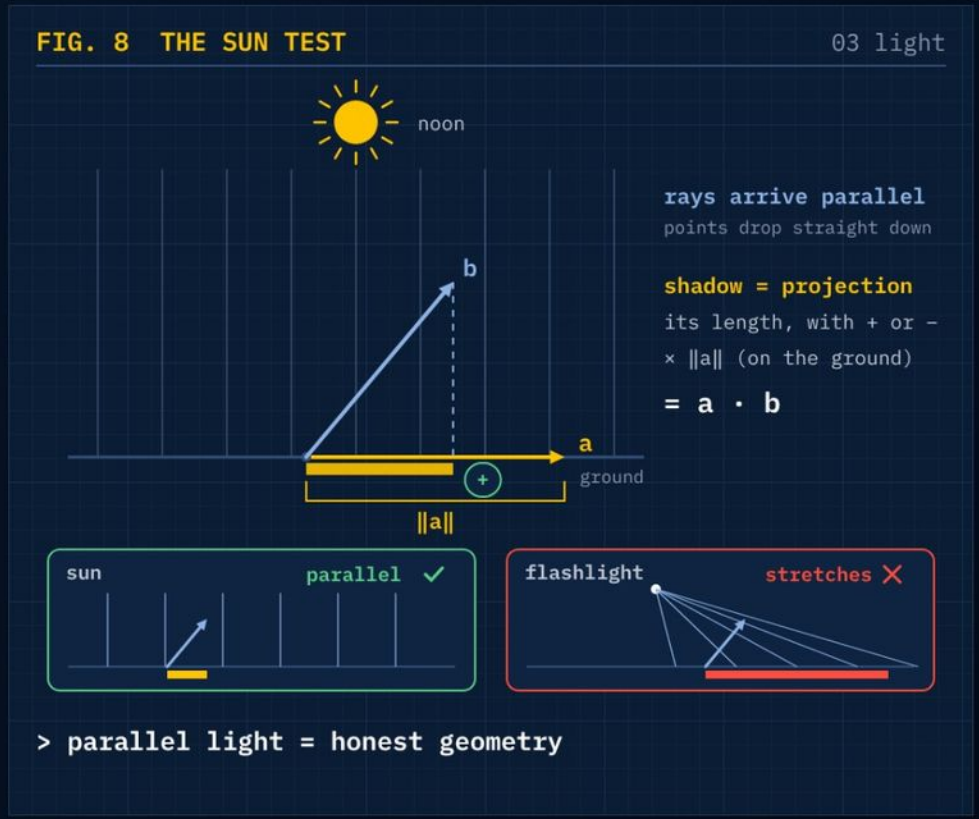
Shadows at noon

Parallel sunlight drops every point straight down.

01 The shadow is the projection

02 Shadow, with its sign, × ‖a‖ = a · b

03 A flashlight would stretch it. The sun never does



Kinvecta analogy, built for this series. Projection idea: 3Blue1Brown, “Dot products and duality,” Essence of linear algebra (2016).

PART 2 · THE NORM

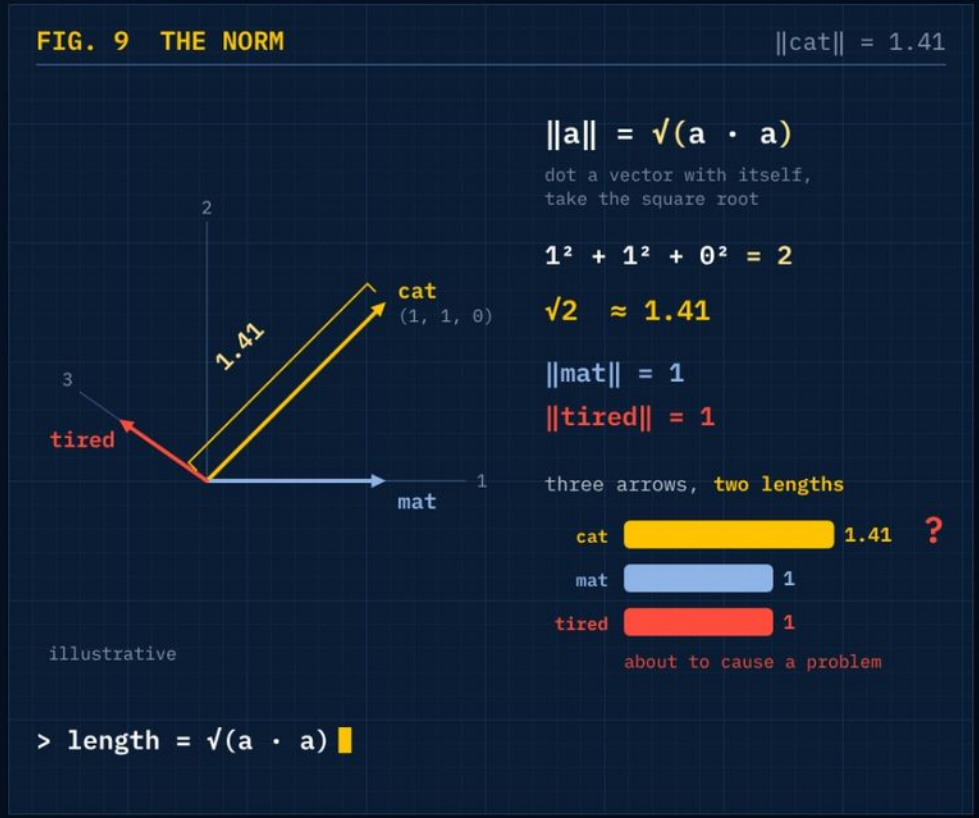
A vector's length

$\|a\| = \sqrt{(a \cdot a)}$

01 $\|cat\| = \sqrt{2} \approx 1.41$

02 $\|mat\| = 1 \cdot \|tired\| = 1$

03 Three arrows, two lengths



Wyss-Gallifent, Math 241 lecture notes §11.3 (Dot Product), University of Maryland. Toy vectors chosen by hand for this episode (axes named by Mark); illustrative, not learned values.

MARK

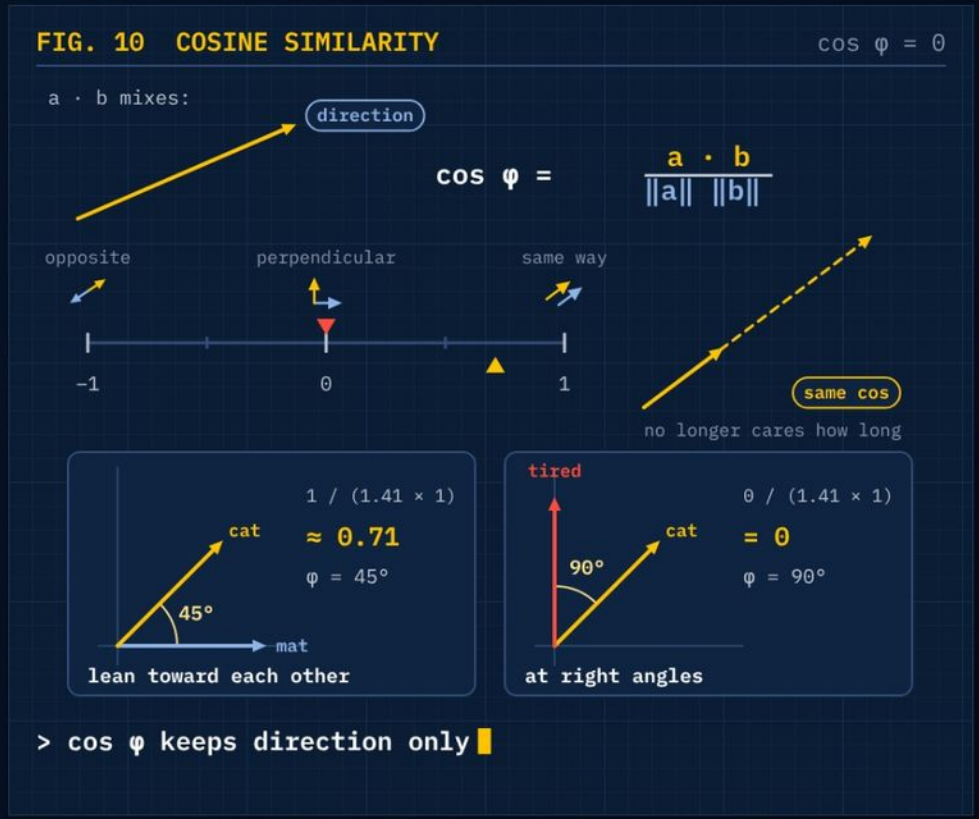
Divide the lengths out

cos φ = a · b / (‖a‖ ‖b‖), always between –1 and 1.

01 It no longer cares how long either arrow is

02 cat and mat: 0.71, a 45° angle

03 cat and tired: 0, a right angle



Turney & Pantel, "From Frequency to Meaning," JAIR 37 (2010). Mikolov, Chen, Corrado & Dean, "Efficient Estimation of Word Representations in Vector Space," arXiv:1301.3781 (2013). Toy vectors chosen by hand for this episode (axes named by Mark); illustrative, not learned values.

PART 3 · THE SCALING TEST

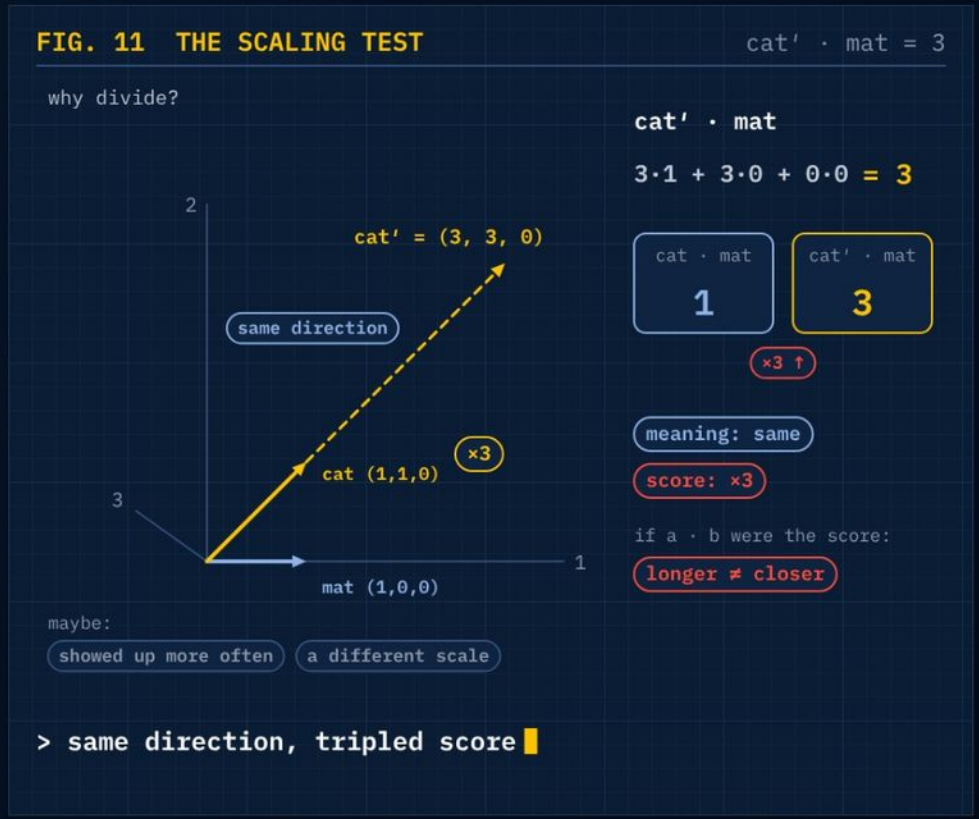
Same direction, three times longer

cat' = (3, 3, 0) points exactly where cat points.

01 cat' · mat = 3. The dot product tripled

02 Nothing about the meaning changed

03 Longer would look like closer



Toy vectors chosen by hand for this episode (axes named by Mark); illustrative, not learned values.

PART 3 · THE SCALING TEST

The cosine didn’t move

3 / (3√2 × 1) ≈ 0.71, exactly what it was.

01 Same direction → same answer

02 Cosine became the standard way to ask “how alike”



Toy vectors chosen by hand for this episode (axes named by Mark); illustrative, not learned values.

PART 3 · WHAT THE LAB SAYS

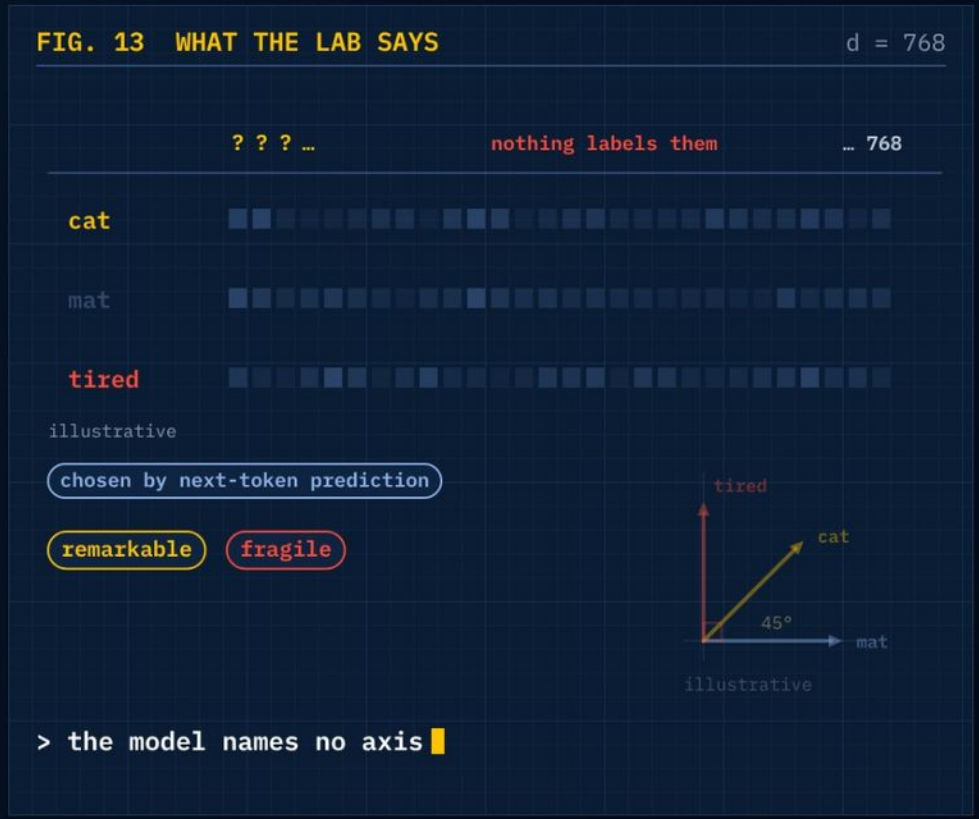
Shared axes, shared direction

cat and mat share axis 1. cat and tired share none.

01 I named the axes. A real model never does

02 It learns a few hundred directions from text

03 Whatever helped predict the next token



Mikolov, Chen, Corrado & Dean, “Efficient Estimation of Word Representations in Vector Space,” arXiv:1301.3781 (2013). Toy vectors chosen by hand for this episode (axes named by Mark); illustrative, not learned values.

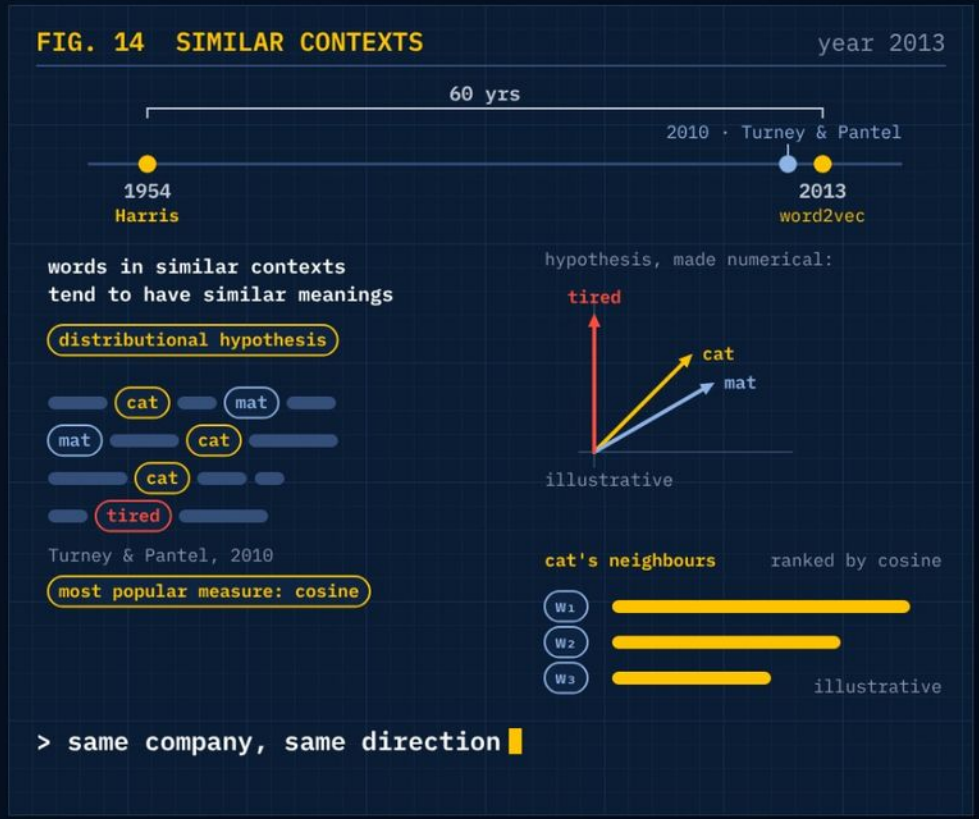
Similar contexts, similar vectors

The distributional hypothesis (Harris, 1954), made numerical.

01 Words in similar contexts tend to have similar meanings

02 Cosine: the most popular similarity measure

03 word2vec ranks nearest neighbours by cosine



Turney & Pantel, "From Frequency to Meaning," JAIR 37 (2010). citing Harris (1954). Mikolov, Chen, Corrado & Dean, "Efficient Estimation of Word Representations in Vector Space," arXiv:1301.3781 (2013).

PART 4 · THE FAMOUS RESULT

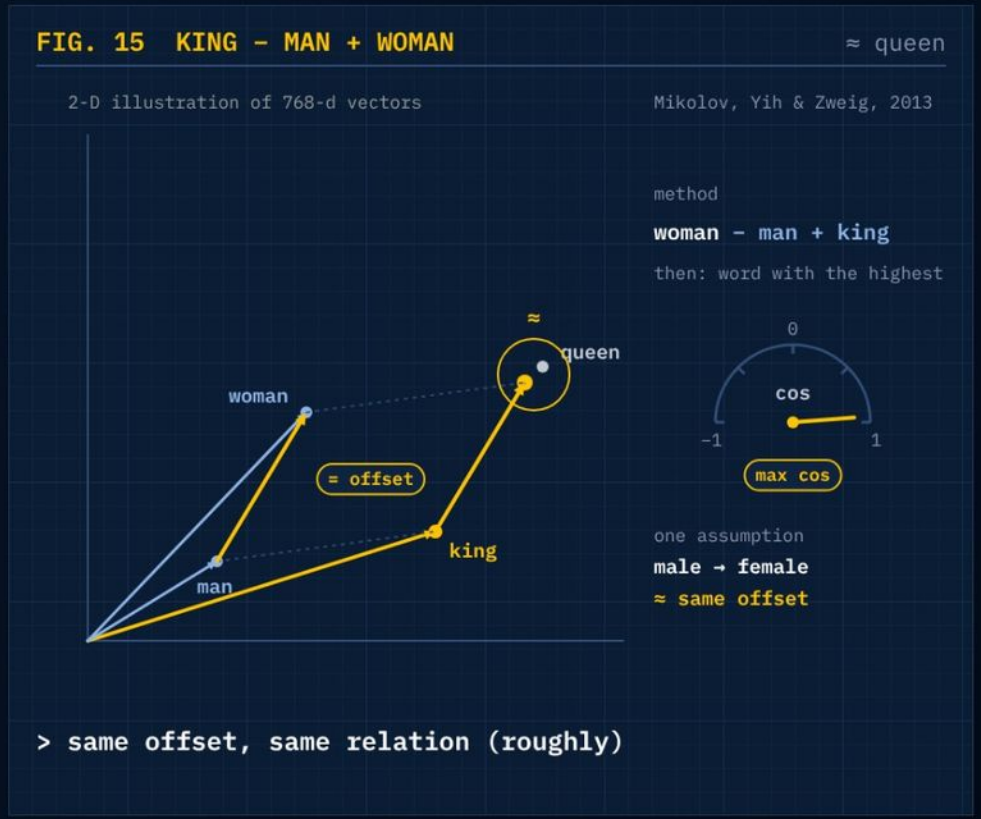
king – man + woman ≈ queen

Mikolov, Yih & Zweig, NAACL 2013.

01 woman – man + king, then the highest cosine

02 Assumes one shared offset per relation

03 A 2-D cartoon of vectors with hundreds of dimensions



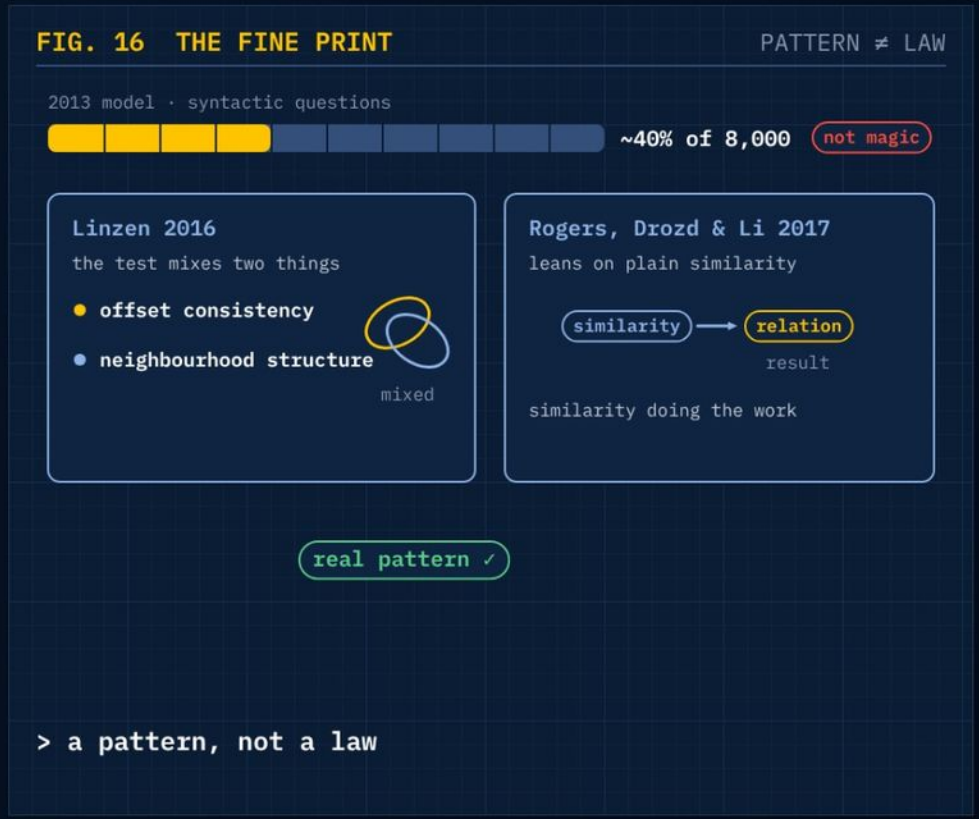
PART 4 · THE FINE PRINT

A pattern, not a law

The 2013 model answered almost 40% of 8,000 syntactic questions.

01 Linzen (2016): the test mixes offsets with neighbourhoods

02 Rogers, Drozd & Li (2017): similarity does part of the work



Mikolov, Yih & Zweig, "Linguistic Regularities in Continuous Space Word Representations," NAACL-HLT 2013. Linzen, "Issues in evaluating semantic spaces using word analogies," RepEval 2016 (arXiv:1606.07736). Rogers, Drozd & Li, "The (too Many) Problems of Analogical Reasoning with Word Vectors," *SEM 2017.

PART 4 · MYTH CHECK

“Close vectors mean similar meaning”

The case for: everything so far. The case against: look inside GPT-2.

01 Ethayarajh (2019) looked inside GPT-2

02 Last layer: two random words, almost perfect cosine

FIG. 17 THE MYTH

cos ≈ 1

"close vectors mean similar meaning"

FOR

- similar contexts
- similar vectors
- cosine finds the neighbours

AGAINST

Ethayarajh 2019 · GPT-2 last layer



> random pairs, near-perfect cosine

Ethayarajh, “How Contextual are Contextualized Word Representations?” EMNLP-IJCNLP 2019. Histogram is illustrative.

PART 4 · WHY IT HAPPENS

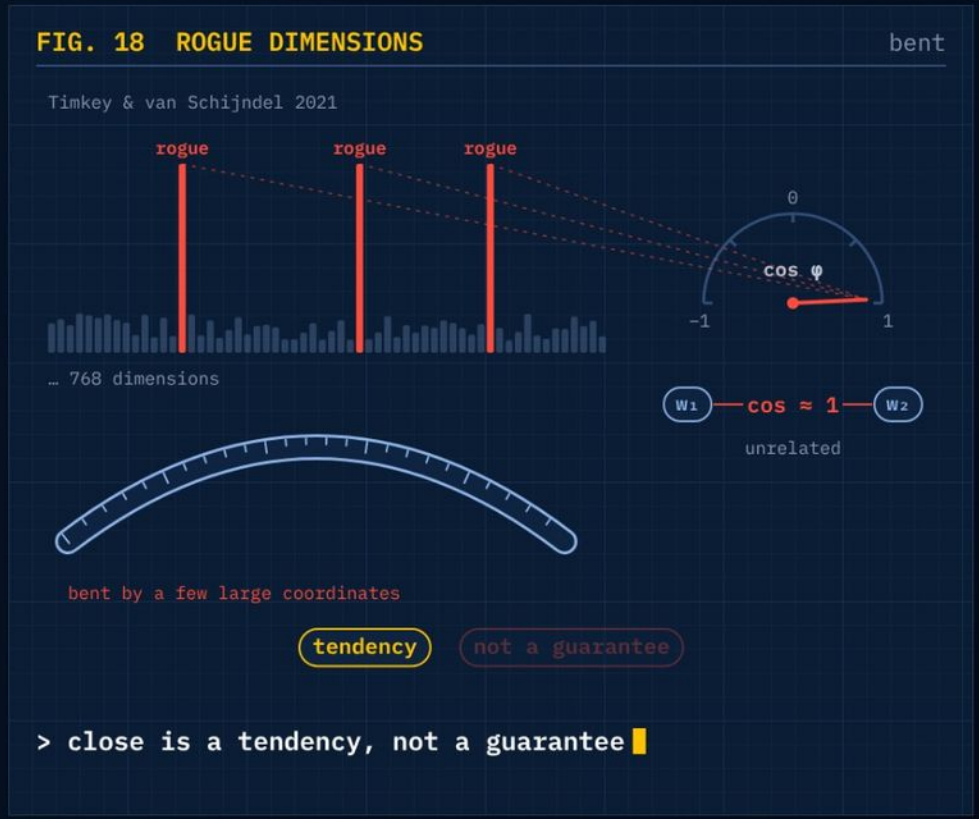
Rogue dimensions

Often just 1 to 3 of 768 coordinates dominate the measure.

01 A few huge coordinates swamp the rest

02 A useful measuring stick that can bend

03 Close is a tendency, not a guarantee



Timkey & van Schijndel, “All Bark and No Bite: Rogue Dimensions in Transformer Language Models,” arXiv:2109.04404 (2021). Bars are illustrative.

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RECAP

Four ideas

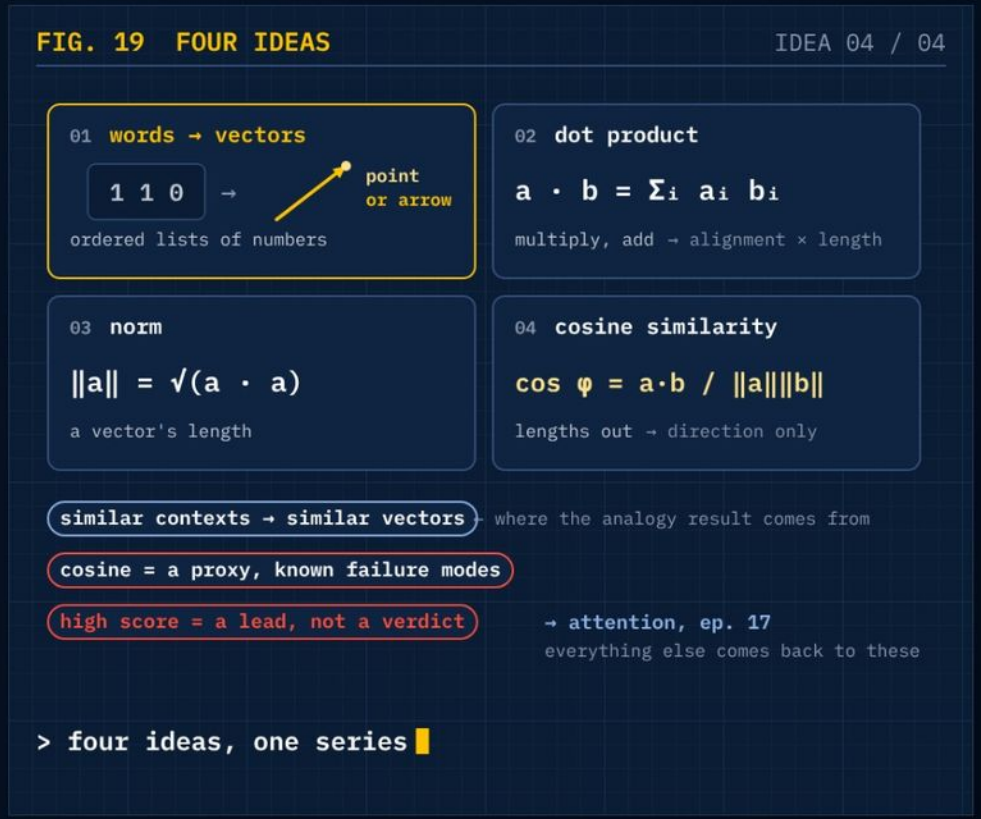
Everything else in the series comes back to these.

01 Words become vectors: lists, points, arrows

02 Dot product: alignment, scaled by length

03 Norm: length. Cosine: direction only

04 A high score is a lead, not a verdict



All sources are listed in the description.

MARK

YOUR TURN

One question to take with you

cat = (1, 1, 0) · mat = (1, 0, 0) ·
tired = (0, 0, 1) · cat′ = (3, 3, 0)

01 Which pair has the larger cosine?

02 Swap cat for cat': which changes, a · b or cos φ?

03 Pause here and work it out



Toy vectors chosen by hand for this episode (axes named by Mark); illustrative, not learned values.

YOUR TURN

The answer

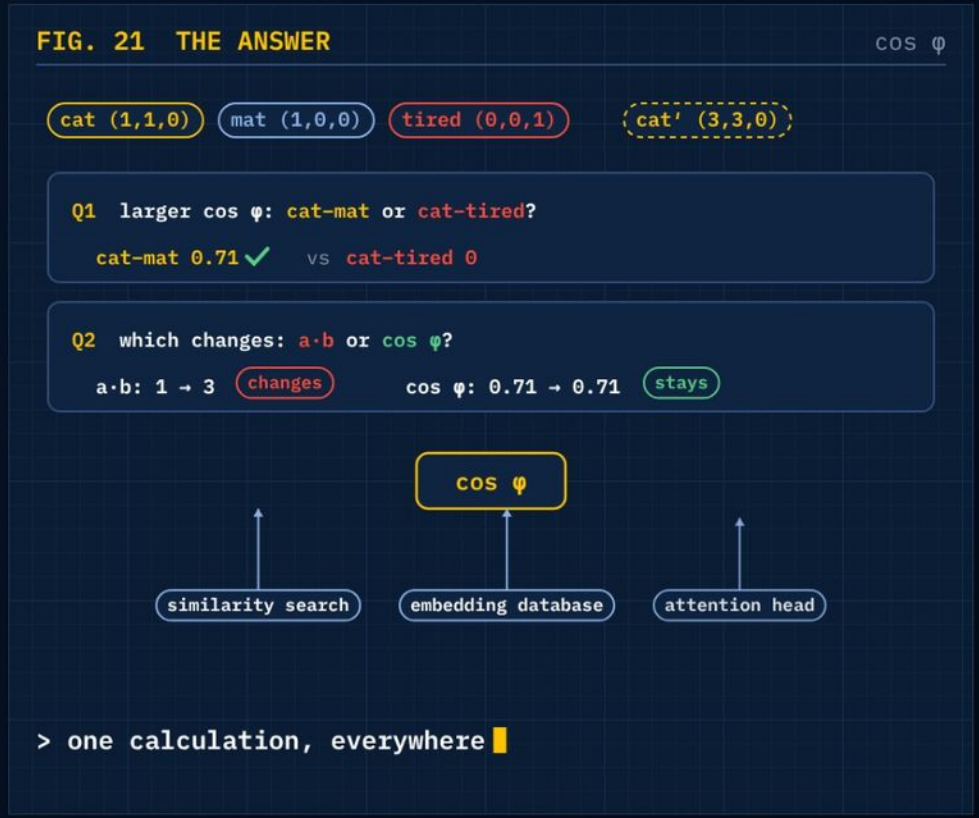
cat–mat wins, and only the dot product moves.

01 cat–mat: 0.71 · cat–tired: 0

02 cat' · mat = 3 (was 1): the dot product changes

03 cos φ stays at 0.71

04 The calculation behind search, embeddings and attention



Toy vectors chosen by hand for this episode (axes named by Mark); illustrative, not learned values.

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TRY THIS WEEK

Three things to try.

01

Do the three dot products on paper

02

Open the notebook and change one axis of cat

03

Give two unrelated words a high cosine

04

Notice how easy that was. That's the lesson

THE MISTAKE TO AVOID

Reading a cosine score as a meaning score. It's an angle. Nothing more.

Companion notebook and every source in the description.

MARK

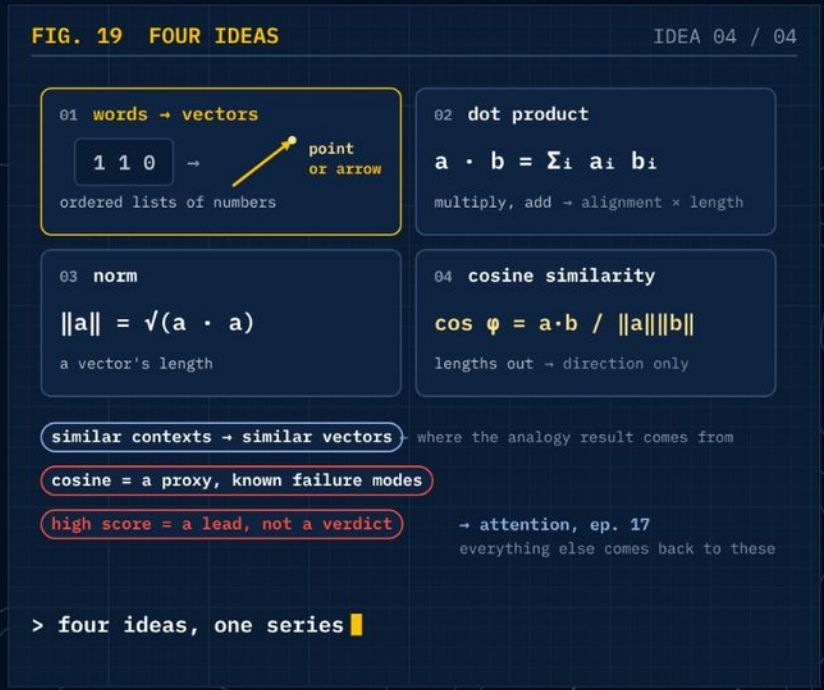
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NEXT TIME

Episode 3: Matrices as machines. Stack the vectors into a table. Numbers go in, transformed numbers come out.

In the comments: which two words would you test first, and what cosine do you expect?

How Large Language Models Work · subscribe for Episode 3



MARK